

Rishi Singh

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EDUCATION

VIT Bhopal University

2023 – 2027

B.Tech in Computer Science and Engineering (AI/ML) – CGPA: 8.5

City Montessori School, Lucknow

2022

Class XII (ISC) – 92.5%

TECHNICAL SKILLS

- **Languages:** Python, Java | **Databases:** SQL, MySQL
- **Frameworks:** FastAPI, Streamlit | **ML Libraries:** TensorFlow, Scikit-Learn, OpenCV, Pandas, SHAP, OR-Tools
- **LLMs & Deep Learning:** Llama-3.1, Groq API, NLP, CNNs, EfficientNet, Transfer Learning, Computer Vision
- **Cloud & Deployment:** Render, Oracle Cloud (OCI), WebSockets, REST APIs, Git, GitHub

PROJECTS

AeroRoster AI: Intelligent Flight & Crew Scheduler

Python, Google OR-Tools, Scikit-Learn, SHAP, Streamlit

- **Links:** [Live Application](#) | [GitHub Repository](#)
- Engineered an enterprise-grade crew scheduling architecture utilizing Google OR-Tools and Constraint Programming to enforce 100% DGCA FDTL legal compliance across flight rosters.
- Integrated a Random Forest machine learning model with Explainable AI (SHAP) to predict and proactively mitigate bio-mathematical crew fatigue, visualizing decision metrics via a Streamlit dashboard.
- Optimized operational efficiency by significantly reducing financial repositioning costs while prioritizing workforce safety.

Dueling Drafts: Real-Time AI Multiplayer Game

Python, FastAPI, WebSockets, Groq API, LLMs (Llama-3.1)

- **Links:** [Live Application](#) | [GitHub Repository](#)
- Architected a real-time asynchronous backend using FastAPI and WebSockets to synchronize live game state and transcript data concurrently across multiple active user sessions.
- Integrated Groq's high-speed inference API with Llama-3.1 to power an automated 'Narrator' and 'Critic', utilizing rigorous regex filters and JSON structuring for robust, crash-proof LLM outputs.
- Implemented comprehensive error handling and fallback routing to ensure seamless user experience during network disconnects or API rate limits, deploying the final containerized service on Render.

Plant Disease Detection & Classifier

Python, TensorFlow, Transfer Learning, Computer Vision

- **Links:** [GitHub Repository](#)
- Engineered a dual-architecture computer vision pipeline to classify 38 distinct agricultural leaf diseases, developing a custom Convolutional Neural Network (CNN) from scratch alongside a fine-tuned EfficientNetB4 model.
- Optimized large-scale data ingestion and preprocessing pipelines utilizing TensorFlow AUTOTUNE and custom augmentation layers, maximizing hardware efficiency and minimizing model convergence time.
- Designed class-mapping logic to export scalable .h5 models, preparing the system for future edge-deployment in diagnostic web and mobile applications.

AWARDS & EXTRACURRICULARS

Club Secretary | Toastmasters International

Sep 2025 – Present

Elected official managing club administration and leading professional development initiatives.

Bhopal, India

CERTIFICATIONS

- **Oracle:** OCI 2025 Certified Generative AI Professional
- **Oracle:** MySQL HeatWave Implementation Certified Associate